The SEERA Dataset: ReadMe File

Name of dataset	SEERA		
Version	1.3		
Total number of attributes/variables	76		
Total number of records	120		
Date of collection	From June 2019 to February 2020		
Donors of dataset	Emtinan I. Mustafa Rasha Osman		
Data collection method	Questionnaire		
Purpose of data collection	• To provide up to date dataset with traditional cost attributes in addition to socio-economic and organizational attributes. • The dataset projects represent constrained technical and economic software development environments. • The dataset fills an urgent gap for the Sudanese and African research community with a more relevant cost estimation dataset that includes factors more aligned with the realities of their software industries. • The SEERA dataset overcomes the current limitations in dataset quality and transparency by augmenting the cost estimation dataset with the original raw data before coding/scaling.		
Country of contribution	Sudan		
Organizations from which data was collected	Types of organizations: • Public and private software development companies • Federal ministries • Federal directorates • Public universities • Freelancers • Corporate IT departments • Telecommunication companies		
Files	Folder	File Name	Description
	Dataset Files\Excel Format	SEERA cost estimation dataset.xlsx	The main 76 attributes and their scores.
		SEERA dataset original raw data.xlsx	Includes the complete set of attributes, sub-attributes, their original values and scores/ratings. Each category of attributes is in separate Excel sheet within the file.
	Dataset Files\ARFF Format	ARFF: SEERA cost estimation dataset.csv.arff	SEERA cost estimation dataset in ARFF format
		ARFF files: SEERA dataset original raw data (sub-folder)	ARFF files for each category of the SEERA dataset original raw data: General Information, Environment, Size, Effort, Users, Developers, Project and Product.
	Additional Files	SEERA dataset attribute formulas.pdf	Includes the formulas used to derive some of the main attributes.

<table>
<tr><td rowspan="6"></td><td rowspan="4"></td><td>SEERA dataset (sub)attribute codes.pdf</td><td>Details the codes corresponding to the options of the categorical/options attributes in the SEERA dataset.</td></tr>
<tr><td>PROMISE2020 Paper.pdf</td><td>The paper discussing the dataset.</td></tr>
<tr><td>Project Programming Languages.pdf</td><td>Distribution of the programming languages implemented within the dataset projects.</td></tr>
<tr><td>PROMISE Paper Tables 7 and 14 and 15.pdf</td><td>Corrections of typos for Tables 7, 14 and 15 in PROMISE paper.</td></tr>
<tr><td rowspan="4">INFSOF Article May 2024 Files</td><td>SEERA cost estimation dataset for prediction.csv</td><td rowspan="4">Files of used for the Random Forest prediction experiments in the research paper: https://doi.org/10.1016/j.infsof.2024.107413</td></tr>
<tr><td>SEERA_Duration SMOGN.csv</td></tr>
<tr><td>SEERA_Effort SMOGN.csv</td></tr>
<tr><td>Data Preparation.pdf</td></tr>
<tr><td>Dataset Format</td><td colspan="3">The dataset is in Excel format and in ARFF format.
Column headings represent attributes/sub-attributes.
Rows represent projects.
Missing values are represented by “?”
The names of sub-attributes not included in the formulas of their main attributes are preceded with a (-).
For projects developed by freelancers: the organization size and department size is encoded as “N/A” and is not considered a missing value.
For values that depend on the answers of conditional questions: if the condition is not met the value is encoded as “N/A” and is not considered a missing value.</td></tr>
<tr><td>Information included within the dataset</td><td colspan="3">ID of organizations that contributed data
Identifiers of projects in each group
Customers type
Start/end date of projects</td></tr>
<tr><td>Dataset information included in PROMISE 2020 research paper</td><td colspan="3">Attributes with outliers
Method/tool used to identify outliers
Number and proportion of incomplete data
List of problems encountered during data collection
Redundant data and reason for redundancy</td></tr>
<tr><td>Dataset information included in the INFSOF 2024 research paper</td><td colspan="3">Dataset statistical characteristics
Comparison of the distribution of project size, effort and duration with the China, SDR and ISBSG10 datasets.
Correlation analysis of the attributes with actual effort and actual duration.</td></tr>
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